

MASON COUNTY AP, WV, USA

WMO: 742079

Lat: 38.915N

Lon: 82.099W

Elev: 196

StdP: 98.99

Time Zone: -5.00 (NAE)

Period: 06-18

WBAN: 63876

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.8	-10.3	-18.7	0.7	-9.7	-16.1	0.9	-8.9	7.8	5.3	7.0	5.5	0.8	280	0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	33.0	22.8	32.4	22.6	31.2	22.2	25.2	30.9	24.5	29.8	23.7	28.6	2.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	18.8	27.8	22.7	17.9	26.9	22.3	17.4	26.3	77.7	30.5	74.9	29.3	71.7	28.9	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.7	5.4	4.7	-16.8	35.2	2.7	1.3	-18.8	36.1	-20.3	36.9	-21.8	37.6	-23.8	38.6		
(5)			-16.6	26.3	3.1	1.5	-18.8	27.4	-20.6	28.3	-22.3	29.1	-24.5	30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.9	0.5	1.8	7.3	13.3	18.8	22.5	23.5	23.3	19.7	13.2	6.9	3.1		
	DBStd	9.54	6.41	6.32	5.54	5.01	4.35	2.92	2.64	2.46	3.82	4.86	4.52	5.26		
	HDD10.0	1043	302	238	120	24	1	0	0	0	22	114	222			
	HDD18.3	2592	553	462	344	162	51	4	1	1	27	172	343	472		
	CDD10.0	2093	7	9	36	123	273	374	418	411	291	121	21	8		
	CDD18.3	600	0	1	2	11	64	128	160	153	68	13	0	0		
	CDH23.3	5977	0	4	26	213	730	1277	1462	1394	697	171	2	1		
	CDH26.7	2269	0	0	5	54	259	500	594	553	261	43	0	0		
(14) Wind	WSAvg	1.3	1.8	1.8	1.7	1.9	1.3	1.1	0.9	0.7	0.8	1.1	1.3	1.5		
(15) Precipitation	PrecAvg	1092	86	74	102	87	120	107	107	92	92	66	73	88		
	PrecMax	1362	159	141	228	157	206	255	138	141	244	185	153	215		
	PrecMin	835	38	21	35	34	14	36	67	32	19	18	36	49		
	PrecStd	147	34	34	44	35	60	49	19	31	61	40	29	40		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.8	22.7	27.0	31.1	32.6	34.1	34.2	35.2	34.0	30.3	23.0	21.1		
		MCWB	14.5	15.2	16.9	17.3	22.2	22.8	24.5	23.1	21.3	20.0	15.1	14.4		
	2%	DB	16.5	18.0	22.6	27.9	31.2	32.6	32.8	32.7	32.2	27.5	21.0	17.7		
		MCWB	13.0	12.6	14.2	16.6	21.4	23.0	23.6	22.5	21.4	18.3	13.3	12.8		
	5%	DB	13.8	15.2	19.9	26.1	29.9	31.3	32.1	32.0	30.0	25.1	17.9	14.9		
		MCWB	11.4	11.5	12.4	15.9	20.6	22.5	23.5	22.5	21.5	17.2	12.2	10.3		
	10%	DB	10.9	12.4	17.0	22.9	27.7	30.0	30.2	30.1	27.5	22.3	16.0	12.1		
		MCWB	8.3	9.1	10.9	14.4	19.4	21.7	22.5	21.8	19.5	16.4	11.1	9.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	17.2	18.0	20.3	24.2	25.7	26.8	25.7	24.6	22.2	16.8	16.2		
		MCDB	17.7	21.4	24.7	25.9	30.0	31.4	32.3	31.2	31.5	25.4	21.2	19.6		
	2%	WB	13.7	14.4	16.1	18.5	22.9	24.7	25.5	24.8	23.3	20.5	14.9	13.7		
		MCDB	16.2	17.5	20.4	24.8	28.9	30.2	31.0	30.2	28.9	25.5	18.4	16.9		
	5%	WB	11.6	12.2	14.4	17.1	21.9	23.8	24.6	23.9	22.3	18.6	13.1	11.5		
		MCDB	13.9	15.0	17.6	22.4	27.5	29.0	29.7	28.9	27.3	23.8	17.1	13.6		
	10%	WB	8.0	8.9	12.3	16.0	20.7	22.9	23.7	23.0	21.2	16.9	11.8	8.8		
		MCDB	10.2	12.0	16.0	21.4	25.6	27.8	28.2	27.6	25.0	21.0	15.1	11.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	10.3	12.0	14.3	13.7	13.1	12.5	13.2	13.7	13.2	12.3	9.2		
		MCDBR	12.7	15.3	16.9	18.7	16.2	15.1	14.2	15.2	15.8	17.0	16.3	14.2		
	5% WB	MCWBR	9.6	10.9	9.3	8.8	7.2	6.5	5.8	6.3	6.6	8.6	10.2	10.2		
		MCWBR	11.6	14.2	12.9	14.5	14.0	12.3	12.1	12.1	13.6	14.8	13.5	12.3		
(40) Clear-Sky Solar Irradiance	taub		0.311	0.325	0.352	0.396	0.441	0.464	0.486	0.485	0.407	0.365	0.330	0.302		
		taud	2.480	2.419	2.405	2.294	2.182	2.164	2.120	2.101	2.301	2.422	2.493	2.529		
	Ebn at Noon		866	898	906	881	844	822	799	789	835	840	835	852		
		Edh at Noon	79	96	108	128	146	149	155	154	119	95	78	71		
(44) All-Sky Solar Radiation	RadAvg		1.75	2.47	3.58	4.80	5.47	6.03	5.71	5.38	4.46	3.17	2.21	1.52		
	RadStd		0.18	0.36	0.30	0.41	0.47	0.36	0.31	0.32	0.43	0.35	0.28	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+126	+65

Nomenclature: See separate page